

⚙️ RS003 Week 2 — English Worksheet

Topic: FPS, Title, and a Colour-Cycling Nebula · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week you think about the code that makes last week's window feel **alive** — a **clock** that controls the frame rate (FPS), a nicer title bar, and a deep-space background that slowly shifts colour like a glowing nebula. You will read it, predict it, fix it, and explain it in your own words.

💬 Words to know: **clock**, **FPS (frames per second)**, **frame**, **tick**, **modulo (%)**

1 · Predict 🧠

Read each piece of code. Just from reading it, write what you think will happen. Do not run it.

```
clock = pygame.time.Clock()
FPS = 60
clock.tick(FPS)
```

What does `clock.tick(60)` do to the speed of the game loop?


```
frame_count = 0
r = (frame_count * 1) % 256
frame_count = frame_count + 1
```

`frame_count` goes up by 1 every loop. What does `% 256` do when it reaches 256?


```
r = (frame_count) % 256
g = 100
b = (200 - frame_count) % 256
screen.fill((r, g, b))
```

Red is climbing and blue is falling, while green stays still. Describe how the nebula colour changes as time passes.


```
pygame.display.set_caption("🌸 Space Shooter v2 - 60 FPS")
```

What part of the window will this line change?

2 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Read it carefully and fix it on paper, then explain why the original was wrong.

Bug A — This game runs fine but your computer's fan gets loud and the CPU hits 100%.

```
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
    screen.fill((100, 100, 200))
    pygame.display.flip()
```

Hint: without a clock, the loop runs as fast as it possibly can.

Write the fixed code (show the loop):

Why was it wrong? Why does your fix work?

Bug B — The nebula should cycle colours, but the program crashes with an error about a colour value being too big.

```
r = frame_count
screen.fill((r, 100, 50))
```

Hint: colour numbers must stay between 0 and 255. `frame_count` keeps growing past 255.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The colour is supposed to keep changing, but it is stuck on one colour forever.

```
frame_count = 0
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
    r = frame_count % 256
    screen.fill((r, 100, 100))
    pygame.display.flip()
    clock.tick(60)
```

Hint: the frame counter never goes up.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this working program line by line. Do not run it. Answer the questions using only what you can see.

```
import pygame
pygame.init()

WIDTH, HEIGHT = 800, 600
screen = pygame.display.set_mode((WIDTH, HEIGHT))
pygame.display.set_caption("🌸 Space Shooter v2 - 60 FPS")

clock = pygame.time.Clock()
FPS = 60
frame_count = 0

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False

    r = (frame_count) % 256
    g = 100
    b = (200 - frame_count) % 256

    screen.fill((r, g, b))
    pygame.display.flip()

    clock.tick(FPS)
    frame_count = frame_count + 1

pygame.quit()
```

Which line decides how many frames happen each second, and what number is it set to?

The green value `g` is always 100. The red and blue values change. Explain why green stays the same colour but red and blue do not.

What does `frame_count = frame_count + 1` do, and why does the nebula need it?

What makes the loop stop and the window close?

`clock.tick(FPS)` is near the bottom of the loop. In your own words, what would happen to the speed of the colours if you removed it?

4 · Explain Your Lesson Code 🎥

Now explain the code **you** wrote during today's lesson. Record a short phone video where you talk through your own nebula code. You can show it running too. Try to use these words: **clock**, **FPS**, **frame**, **tick**, **modulo**.

1. Show your background slowly changing colour.
2. Read your colour formula out loud and say what makes it change.
3. Say what FPS you used and what `clock.tick` does for your computer.
4. Point to the line that makes the colour keep moving.

Write what you will say in your video. Plan it here before you record.

Submit 

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.